

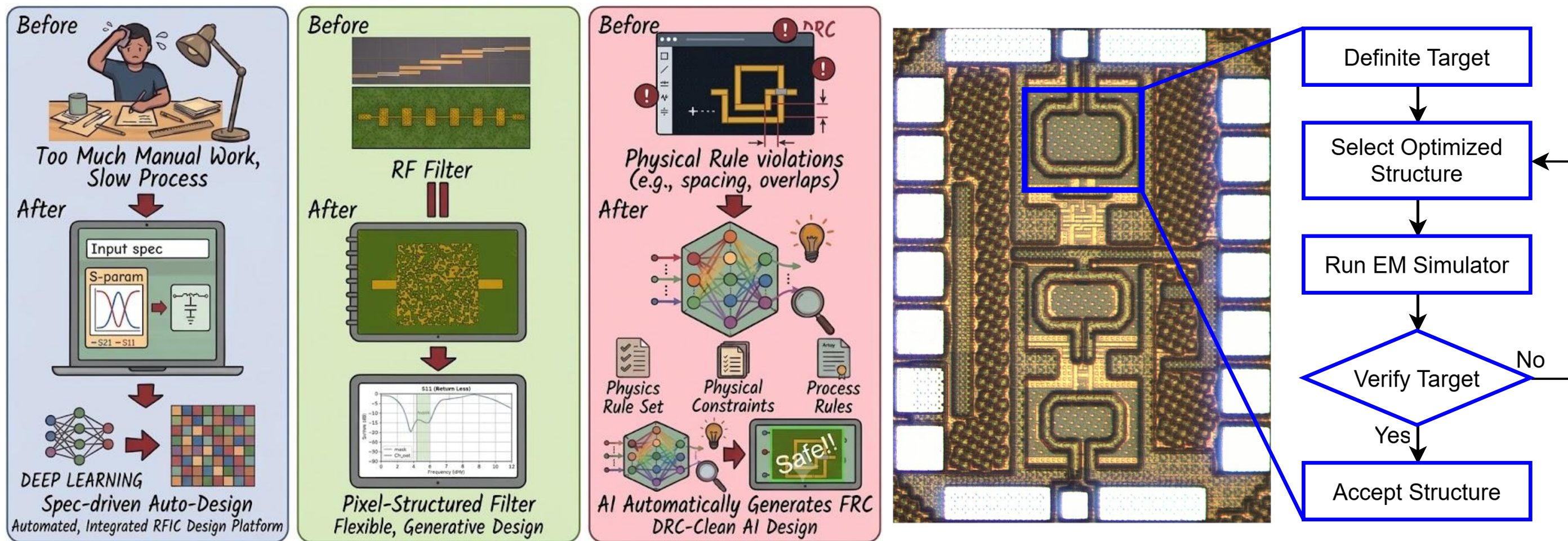
Physics-Aware AI-Driven Pixelated RF Band-Pass Filter Circuits Inverse Design

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1 Motivation

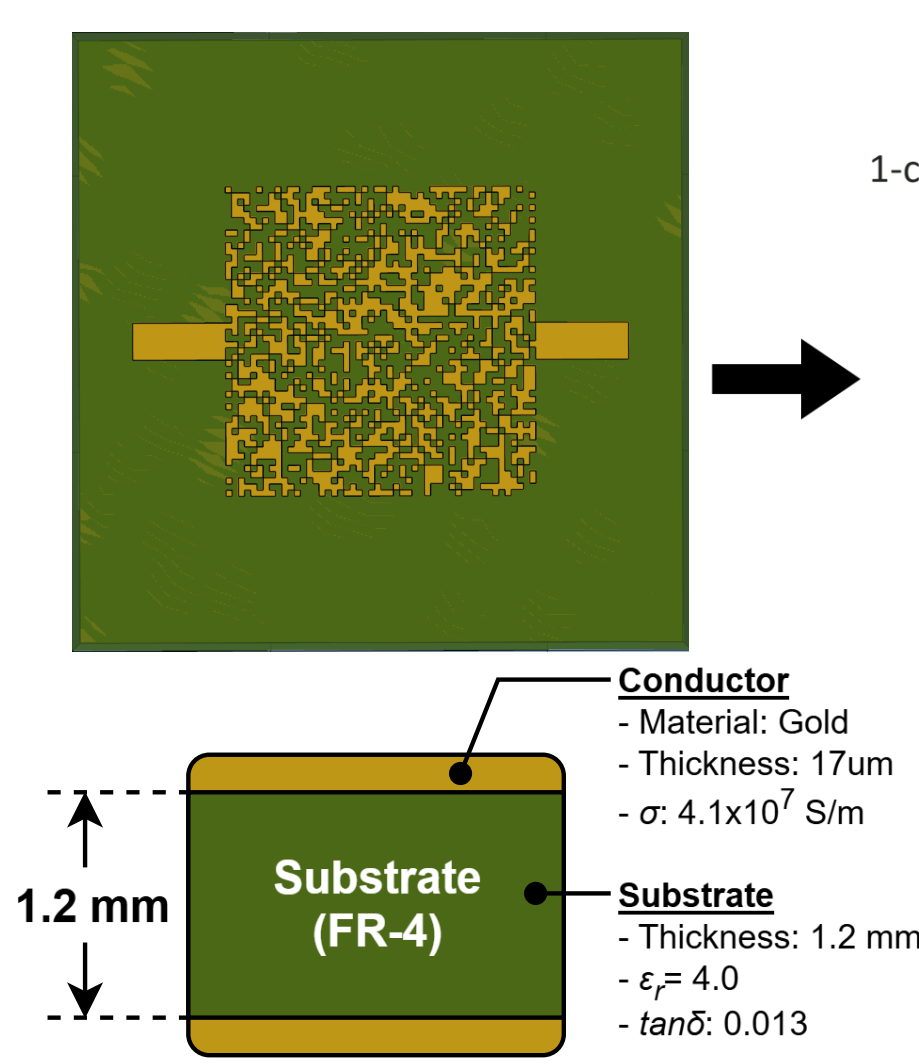
Why Automated Inverse Design?



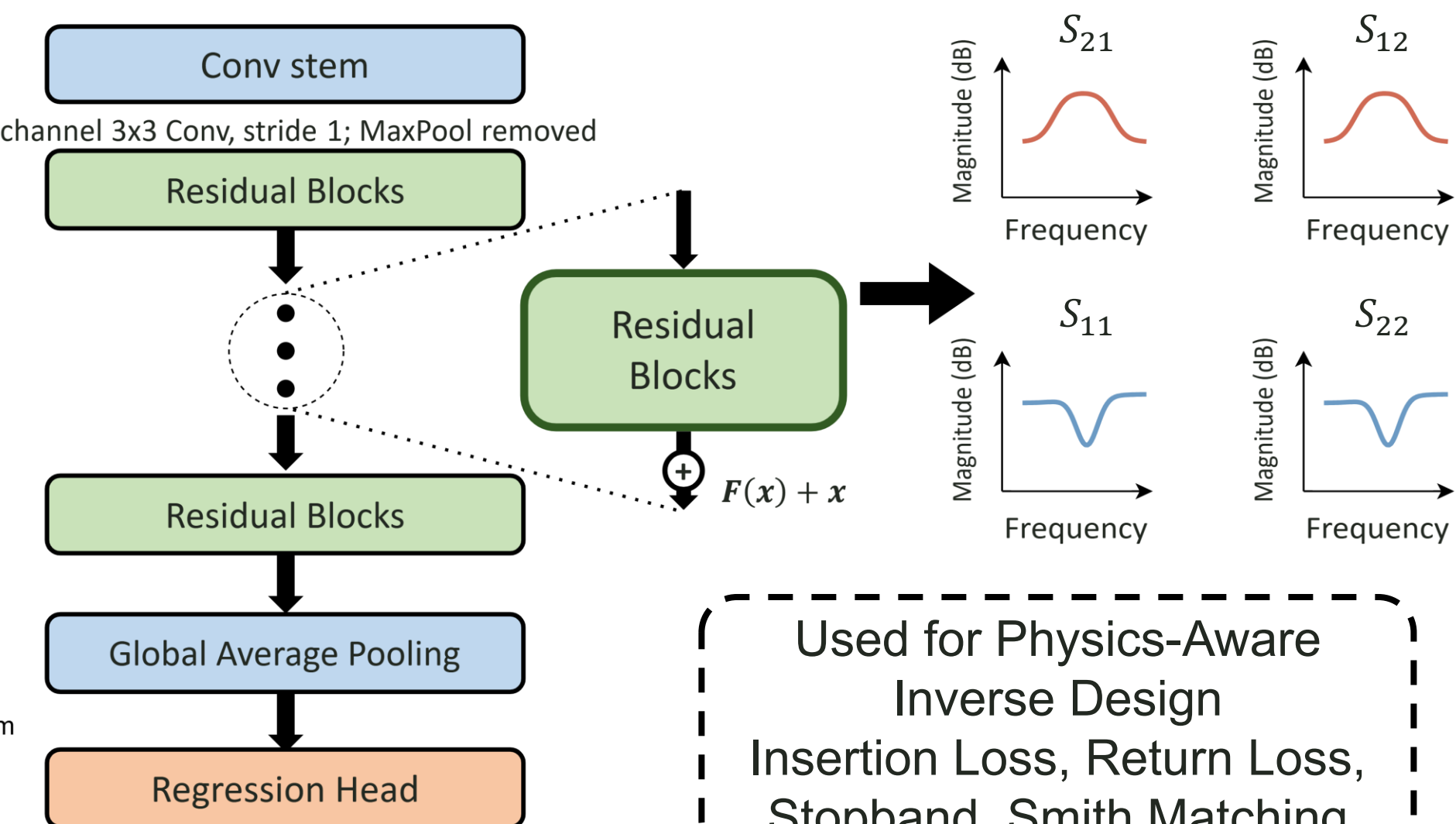
- ✓ **Design Inefficiency of Conventional Manual RF Filter Design**
- ✓ **Limitations of Fixed Template-Based Design Approaches**
 - Restricted design area compact, non-intuitive structures
- ✓ **Simulation Bottleneck in Current AI-Assisted Design Methods**
 - Hundreds of EM simulations, repeated for every new spec

3 Generation of Random Pixel Structures and Learning

Layout & Substrate



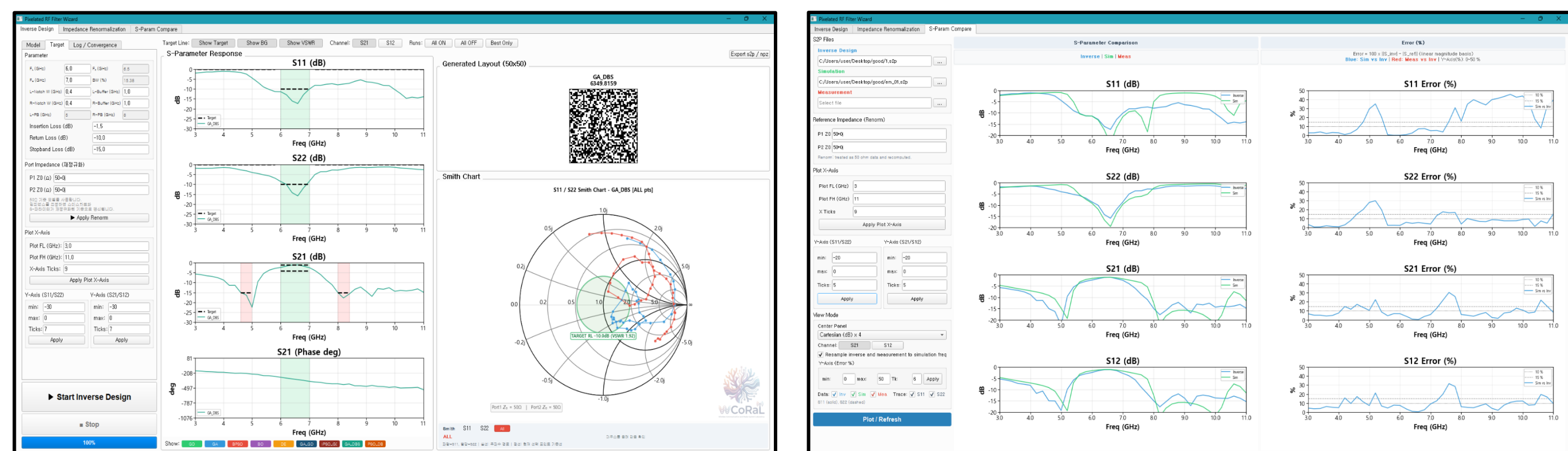
ResNet-based Learning



- ✓ **PCB: 20 mm x 20 mm and Pixel: 10 mm x 10 mm, 50*50 pixel.**
- ✓ **50x50 binary pixelated RF BPF layout is used as a 2D image input.**
- ✓ **ResNet-based surrogate model predicts S-parameters.**

5 Surrogate-Driven GA-DBS Inverse Design

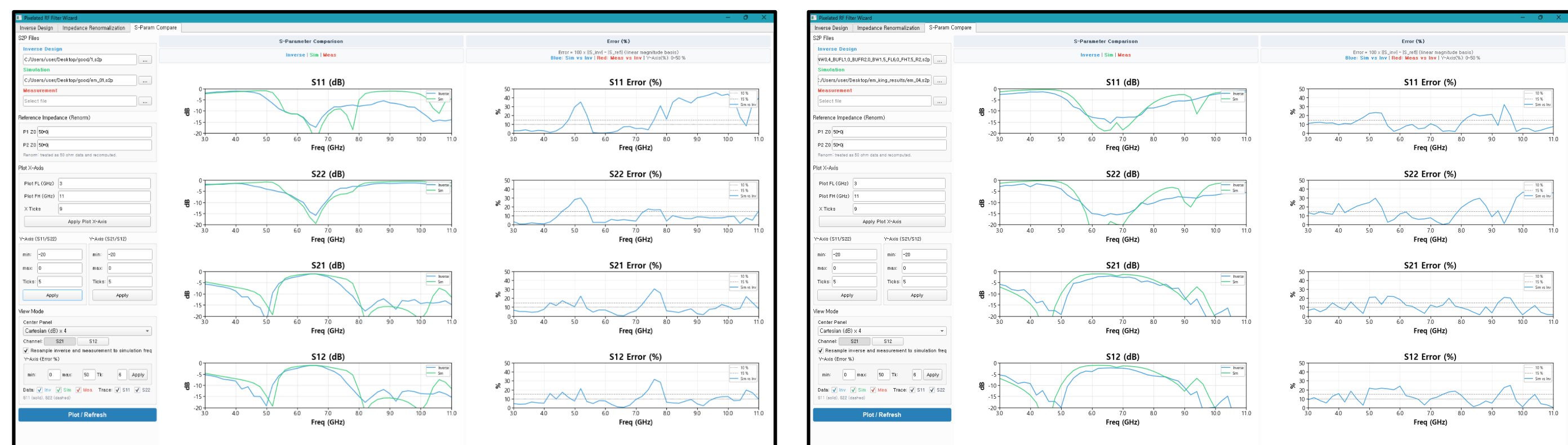
Interactive GUI for Pixel Layout Prediction and Evaluation



- ✓ **User-defined RF specifications can be directly entered.**
- ✓ **GA-DBS optimization automatically searches for binary pixel layouts that satisfy the specified RF targets.**
- ✓ **Generated layouts and predicted S-parameters are visualized in real time.**
- ✓ **Smith chart and S2P/NPZ export functions for EM verification.**
- ✓ **Compared with prediction results and EM simulation results to error rate.**

7 Prediction vs EM Simulation Relative Error Rate

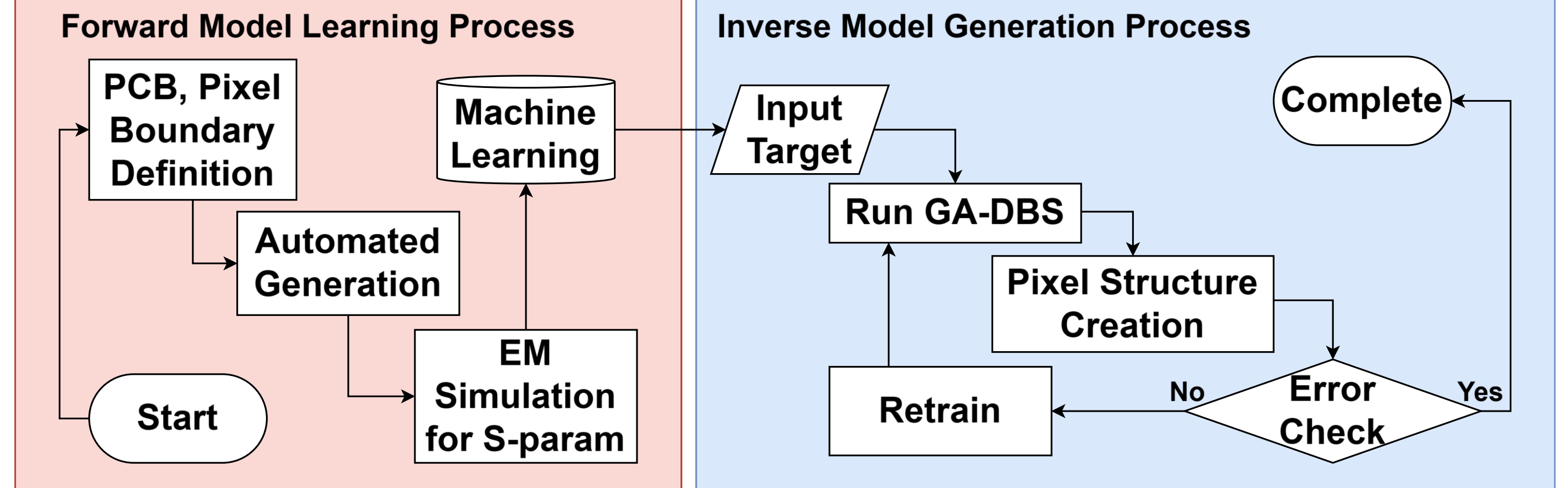
EM Simulated S-Parameter and Relative Error Rate



- ✓ **Full-wave EM validation was performed on two inverse design.**
- ✓ **Verify the surrogate-predicted S-parameter responses.**
 - BPF Model 1 @ 6.0 – 7.0 GHz
Insertion Loss Error: $\leq 9.0\%FS$ and Return Loss Error: $\leq 7.5\%FS$
 - BPF Model 2 @ 5.5 – 7.7 GHz
Insertion Loss Error: $\leq 22.2\%FS$ and Return Loss Error: $\leq 14.1\%FS$

2 AI-based Inverse Design Framework

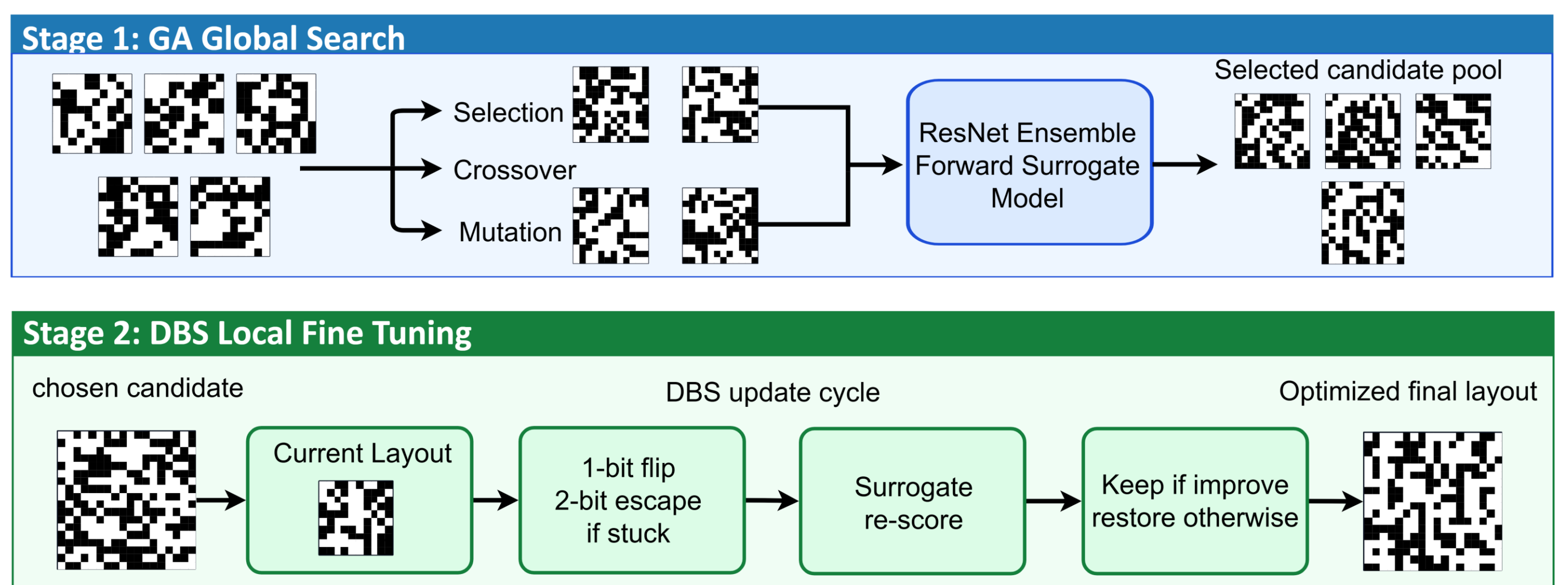
Learning – Generation Process



- ✓ **Dataset Construction from Random Pixel Layouts**
 - $N \times N$ binary maps labeled by EM simulation for S-parameters
- ✓ **PINN/ResNet Surrogate Replacing the Conventional EM Solver**
 - Near-instant evaluation, reusable across different target specifications
- ✓ **GA-DBS Inverse Search with Verification Loop**
 - Candidate layouts refined by error check and retraining until convergence

4 ResNet-based Ensemble Surrogate Model

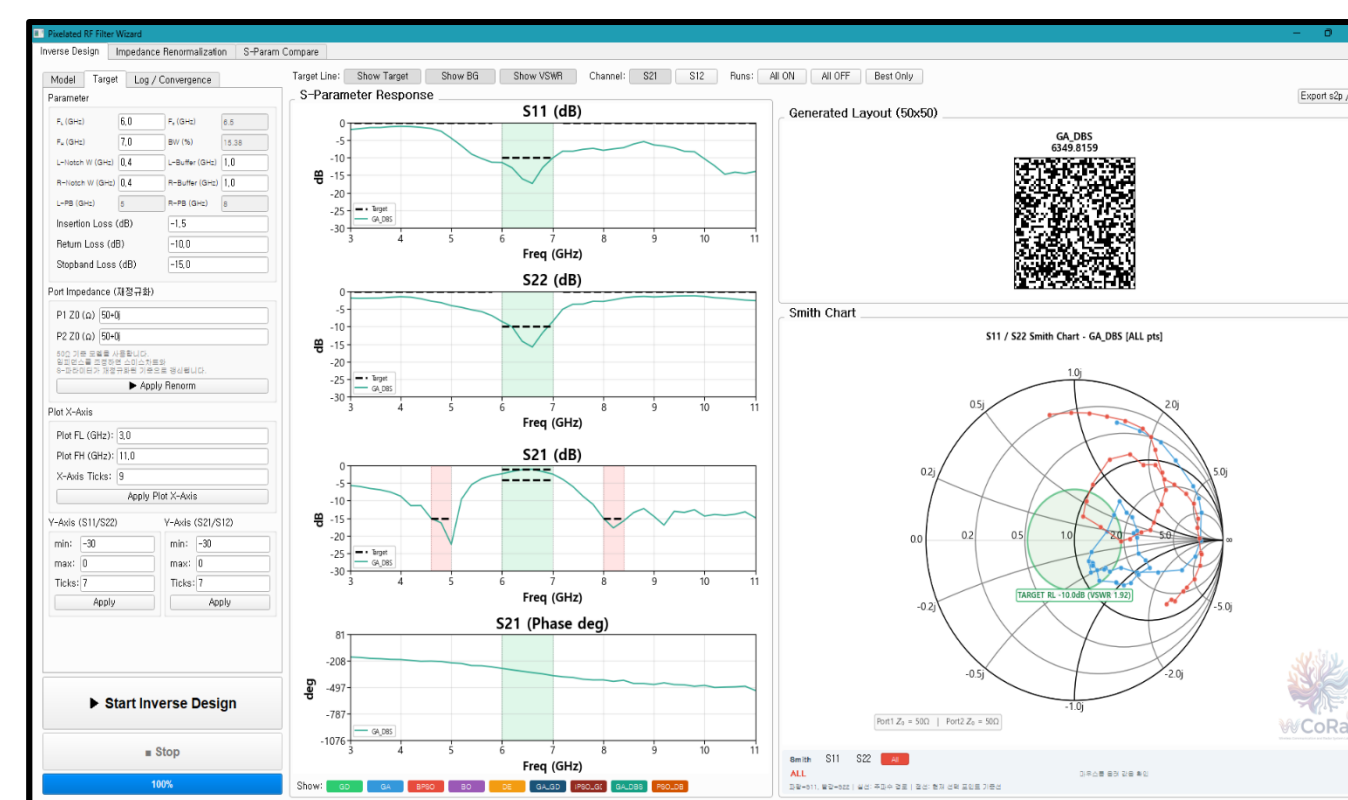
GA-DBS Surrogate Optimization



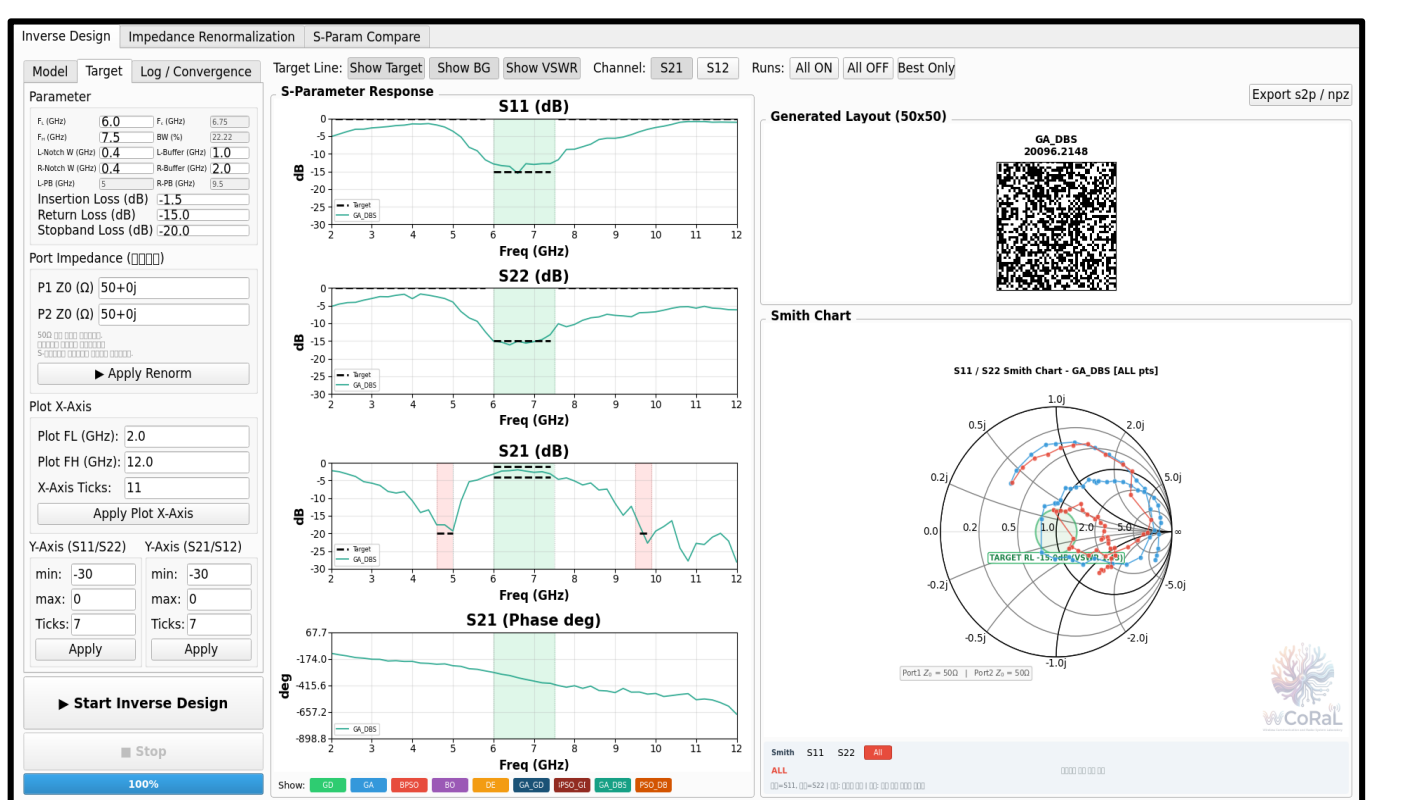
- ✓ **GA explores diverse binary pixel candidates globally.**
- ✓ **DBS refines the selected layout through local pixel flipping.**
- ✓ **This two-stage strategy improves search efficiency while enabling pixel-level refinement and reducing local-optimum risk.**

6 AI-Based RF Filter Inverse Design

BPF model 1 (Narrowband)



BPF model 2 (Wideband)



- ✓ **BPF Model 1 @ 6.0 – 7.0 GHz with 1 GHz Bandwidth**
 - Insertion Loss: -1.1 dB @ 6.5 GHz and Return Loss ≤ -10.0 dB.
 - Roll-off rate: 22.3 dB/GHz @ low band and 16.9 dB/GHz @ high band
- ✓ **BPF Model 2 @ 5.5 – 7.7 GHz with 2.2 GHz Bandwidth**
 - Insertion Loss: -1.3 dB @ 6.6 GHz and Return Loss ≤ -10.0 dB.
 - Roll-off rate: 25.1 dB/GHz @ low band and 10.6 dB/GHz @ high band

8 Summary and Conclusion

Summary

	BPF Model 1	BPF Model 2
Center Frequency	6.5 GHz	6.6 GHz
Bandwidth _{RL-10dB}	6.0 GHz – 7.0 GHz	5.5 GHz – 7.7 GHz
Peak Insertion Loss	-1.1 dB @ 6.5 GHz	-1.3 dB @ 6.6 GHz
Return Loss	≤ -10.0 dB @ BW	≤ -10.0 dB @ BW
Roll-off Factor	≥ 16.9 dB/GHz	≥ 10.6 dB/GHz
RMS Error @ IL	$\leq 9.0\%FS$ @ BW	$\leq 22.2\%FS$ @ BW
RMS Error @ RL	$\leq 7.5\%FS$ @ BW	$\leq 14.1\%FS$ @ BW

- ✓ **The proposed inverse design framework can generate the desired RF BPF.**
 - Using the GA-DBS optimization process, the desired RF filter layout was designed within ~10 minutes.
 - This work presents an innovative approach to transforming the conventional manual RF filter design paradigm by integrating learning, inverse design, EM simulation, and optimization within an open-source framework.